

JGR Planets

RESEARCH ARTICLE

10.1029/2022JE007510

Key Points:

- The crater population in equilibrium can be combined with a production model to directly constrain crater lifetime for <~200-m diameter craters
- With known initial crater shapes and erasure thresholds, the effective diffusivity compatible with these crater lifetimes can be inferred
- If volatiles are found in small (<4-m) craters and were delivered after crater formation, the volatiles are <~50 Ma old

Supporting Information:

Supporting Information may be found in the online version of this article.

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Citation:

Fassett, C. I., Beyer, R. A., Deutsch, A. N., Hirabayashi, M., Leight, C. J., Mahanti, P., et al. (2022). Topographic diffusion revisited: Small crater lifetime on the Moon and implications for volatile exploration. *Journal of Geophysical Research: Planets*, 127, e2022JE007510. <https://doi.org/10.1029/2022JE007510>

Received 28 JUL 2022
Accepted 17 NOV 2022

Topographic Diffusion Revisited: Small Crater Lifetime on the Moon and Implications for Volatile Exploration

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Abstract Crater degradation and erosion control the lifetime of craters in the meter-to-kilometer diameter range on the lunar surface. A consequence of this crater degradation process is that meter-scale craters survive for a comparatively short time on the lunar surface in geologic terms. Here, we derive crater lifetimes for craters of <~200 m in diameter by analyzing existing functional expressions for crater population equilibrium and production. These lifetimes allow us to constrain the topographic degradation needed at different scales to explain when craters become undetectable on equilibrium surfaces. We show how topographic degradation can be treated as a process of anomalous (scale-dependent) topographic diffusion and find large differences in effective diffusivities at different scales, consistent with a wide range of evidence besides equilibrium behavior. Understanding the range of morphology of meter-scale craters is particularly relevant for future exploration of the lunar surface with rovers. We illustrate expectations for the d/D distribution of small lunar craters on surfaces with negligible regional-scale slopes. Our results imply that if volatiles are found in preserved <4 m craters and were delivered after crater formation, the volatiles must have been emplaced in the last ~50 Ma. Given the rates of surface evolution we infer, the most likely emplacement time for any volatiles discovered at or near the surface in the interior of fresh, small craters may be much younger than this upper limit.

Plain Language Summary Impact craters are the most common landform on the Moon and form across a range of sizes and at vastly different rates. Small, meter-sized craters form much more frequently than large, kilometer-sized craters. After crater formation, craters start to widen, fill in, and their topographic relief is reduced due to their exposure to the lunar environment. Eventually, this infilling process is sufficient to render craters unrecognizable on the surface. The time over which this occurs is the crater lifetime, which is a function of crater diameter. This paper quantifies the crater lifetime. As suggested by some earlier workers, we show that, for 10–100 m craters, lifetime increases as a function of diameter to a power p of ~1.1–1.3. This diverges from what would be expected for one model for how regolith on the Moon is transported (“classical diffusion”), which would have $p = 2$, and supports a different model (“anomalous diffusion”) that implies the rate of infilling depends on the scale being considered. In this study, we also show that this model for surface evolution implies that any volatiles that end up being discovered within small fresh craters on the Moon must have gotten there recently.

1. Introduction

This paper revisits an element of the crater degradation analysis in Fassett and Thomson (2014) (hereafter FT2014), specifically the lifetime of lunar craters in the meter to kilometer diameter size range. New theoretical understanding, observations, and analyses prompt a revision to the extrapolated small crater lifetimes quoted in FT2014 for $D < 800$ m craters. In particular, topographic change on the Moon is scale-dependent in a manner not recognized by FT2014 (e.g., Du et al., 2019; Fassett et al., 2018; Minton et al., 2019; Xie et al., 2017). This necessitates reconsideration of how diffusive degradation of sub-kilometer-scale craters is treated. We describe a straightforward approach for calculating crater lifetimes when crater populations are in equilibrium, where craters are being removed as fast as they form on average (e.g., Gault, 1970). In doing so, we also show how the rate of geomorphic change at different scales—parameterized as a size-dependent diffusivity—can be estimated if both the initial crater morphometry and threshold for when craters become undetectable on equilibrium surfaces are

known. Comparing crater production and erasure rates controlling equilibrium allows for determination of crater lifetime and degradation down to meter-scales, which are especially relevant to understand for exploration.

Impact cratering dominates the geomorphic evolution of the Moon and is the dominant factor in its topographic evolution because of its ubiquity. A key element of the accumulation of a crater population is that small craters are much more common than large ones, because there is an approximate power-law distribution of crater production for sub-km sizes, $N_{\geq D} = cD^{\beta}$, that has an exponent of β of about -3.1 (based on the Neukum Production Function between $D = 10$ and 100 m; Neukum et al., 2001). Since the power-law slope of this cumulative distribution is steeper than -2 ($\beta < -2$), small lunar craters ($D < \sim 200$ m) cumulatively affect more surface area than large craters (~ 1 km and larger) (e.g., Melosh, 1989). For this reason, many workers (including FT2014) considered the geomorphic forcing induced by these small craters as a continuum process where the net effect on the landscape and individual landforms is diffusive (Craddock & Howard, 2000; Howard, 2007; Ross, 1968; Soderblom, 1970; Xie et al., 2017). However, Minton et al. (2019) argued that small primary craters may not be the primary driver of topographic evolution, since secondary cratering and disturbances triggered by distal ejecta affect areas much larger than primaries. The detailed forcing of topographic diffusion on the Moon remains imperfectly understood.

Topographic diffusion has been widely applied to describe terrestrial hillslope evolution (e.g., Culling, 1960; Pelletier, 2008). For a topography field h , with topographic diffusivity κ (with units of length²/time), classical diffusive evolution is written as

$$\frac{\partial h}{\partial t} = \kappa \nabla^2 h \quad (1)$$

If this equation fully described crater evolution, and diffusivity and crater shape were independent of scale, then the diffusive crater lifetime for diameter D , $\tau(D)$, would scale as $\tau(D) \sim D^2$, as is standard for Fickian diffusion. This lifetime represents the average period that a crater will survive as recognizable in the surface population. The extrapolated crater lifetimes in FT2014 were based on this idea that crater lifetimes scale as $\tau(D) \sim D^2$, but this turns out to be a flawed assumption, because the effective diffusivity κ must vary with scale given a variety of observational and numerical constraints (Fassett et al., 2018; Li & Mustard, 2000; Minton et al., 2019; Xie et al., 2017; Zhang et al., 2021). Here, we focus on how crater lifetime and size-dependent topographic diffusivity must vary to be consistent with the observed crater equilibrium populations on the Moon.

2. Equilibrium and Crater Degradation

A population of craters (at some crater size) is said to be in equilibrium if production of craters (at that size) is in balance with removal of craters (at that size) from the population (Gault, 1970). Because of the steep production function for sub-km craters, smaller craters reach equilibrium faster than larger craters, so craters below some threshold diameter are all in equilibrium. This threshold diameter increases in size with time: on the lunar mare, equilibrium is observed to occur on virtually all surface units for craters smaller than $D \sim 100$ – 200 m, with older surfaces having larger craters in equilibrium (Gault, 1970; Xiao & Werner, 2015). In this state, some information about impact history is lost, but the equilibrium size-frequency distribution of craters also provides a powerful constraint on crater degradation and obliteration (Hirabayashi et al., 2018; Minton et al., 2019; Shoemaker, 1965).

With a <1 km production function proportional to $D^{-3.1}$ as mentioned earlier, if the crater removal rate (inverse of τ) was driven by standard diffusion so crater lifetime was proportional to D^{-2} , the equilibrium size-frequency distribution that arises would have a slope of approximately $-3.1 - (-2) = -1.1$ (i.e., production—removal; Minton et al., 2019, see their Section 2.2 and Equation 18; see also Zhang et al., 2021, Equation 3). However, a -1.1 slope is much shallower than the equilibrium power-law slope for sub-kilometer craters that is observed on plains surfaces, which have a power law slope of -1.8 to -2 (see Equation 3, below). To achieve consistency with the balance of crater production and removal, all else being equal, the observed equilibrium size-frequency distribution would require craters to have diffusive lifetimes τ with a size dependence of $\tau \sim D^{1.1} - \tau \sim D^{1.3}$. The scaling of crater lifetimes in equilibrium thus requires anomalous diffusion, as also implicated by other data (Du et al., 2019; Fassett et al., 2018; Li & Mustard, 2000; Minton et al., 2019; Xie et al., 2017).

3. Calculating Crater Lifetimes in Equilibrium

The balance of crater production and destruction in equilibrium controls both the equilibrium population behavior (cf. Xiao & Werner, 2015) and crater lifetime (or retention/detectability period) at a given crater size. In other words, if we know the production rate $\Phi(D, T)$ of craters and the number of craters typically observed in equilibrium, $N_{eq}(D)$, the diffusive lifetime of craters, $\tau(D)$, can then be directly calculated, in an average sense:

$$\tau(D) = N_{eq}(D) / \Phi(D, T) \quad (2)$$

Note that $N_{eq}(D)$ and $\Phi(D, T)$ are commonly expressed as a cumulative number (or rate) greater than diameter D in size, though these functions can be equivalently expressed in a differential form.

For the Moon, established models for both $N_{eq}(D)$ and $\Phi(D)$ exist, at least for craters $D \geq 10$ m. Two commonly used models for $N_{eq}(D)$ are from (a) Trask (1966) (sometimes called “standard lunar equilibrium”), and (b) Hartmann (1984):

$$N_{eq_Trask}(D) = 10^{-1.1} D^{-2} \quad (3a)$$

$$N_{eq_Hartmann}(D) = 10^{-1.14} D^{-1.83} \quad (3b)$$

D here is in kilometers, and $N_{eq}(D)$ is the cumulative number greater than D per km².

For the production rate of craters at a given size, the most widely used model comes from Neukum (1983) or a later iteration, Neukum et al. (2001). The strong agreement of the measured ages on samples returned by Chang'E 5 with expectations predicted by the Neukum model demonstrates its usefulness (Che et al., 2021; Li et al., 2021; Yue et al., 2022). The modern $D \sim 10$ m flux also appears to be in strong agreement with the Neukum flux, falling within $\sim 20\%$ – 30% (Speyerer et al., 2016).

The Neukum model is defined in terms of a *chronology function* (Equations 4 and 5 below) and *production function* (Equation 6 below) (see Neukum, 1983). An assumption of the model is that the relative size of impactors averaged over geologically meaningful periods remains consistent with time. If this assumption holds, then crater formation as a function of time and as a function of size are independent, and can be defined separately, with the chronology function capturing all variations of the crater formation rate and the production function describing the relative formation rate of different sized craters with respect to a reference size (see, e.g., C. I. Fassett, 2016 for review and additional discussion). Neukum (1983)'s chronology function $C_{\geq 1km}(T)$, gives the frequency of ≥ 1 km craters formed per km² since age T in Ga with

$$C_{\geq 1km}(T) = 5.44 \times 10^{-14} (e^{6.93T} - 1) + 8.38 \times 10^{-4} T \quad (4)$$

The differential form of this equation gives the instantaneous formation rate of ≥ 1 -km craters:

$$dC_{\geq 1km}/dT = 3.77 \times 10^{-13} e^{6.93T} + 8.38 \times 10^{-4} \quad (5)$$

For ages less than 2.5 Ga, the exponential term is insignificant, and the formation rate of craters is constant, $dC_{\geq 1km}/dT = 8.38 \times 10^{-4}$ craters per km² per Ga. In other words, over the last half of lunar history, for every million square kilometers, ~ 838 craters $D \geq 1$ km are expected to form per billion years.

The production function gives the relationship for the formation rate of craters of any size, $\geq D$, as a function of the formation rate of craters at a reference size $D \geq 1$ km. The definition for the production function differed in the 1983 and 2001 versions of Neukum's model, but in each instance the definition was given as a summed polynomial function relating the $\log_{10}$ of the cumulative crater density $\Phi(D, 1 \text{ Ga})$ to the $\log_{10}$ of the diameter (in km):

$$\log_{10} \Phi(D, 1 \text{ Ga}) = \sum_{i=0}^{i=11} a_i (\log_{10} D)^i \quad (6)$$

The a_i coefficients for both the Neukum (1983) and Neukum et al. (2001) versions of this function are reproduced in Table S1 in Supporting Information S1. For calculations, it is often convenient to normalize this so that for $D \geq 1$ km, the function expresses a ratio Θ of the number of craters $\geq D$ relative to $D \geq 1$ km:

$$\Theta(D) = (10^{\sum_{i=0}^{i=11} a_i (\log_{10} D)^i}) / 8.38 \times 10^{-4} \quad (7)$$

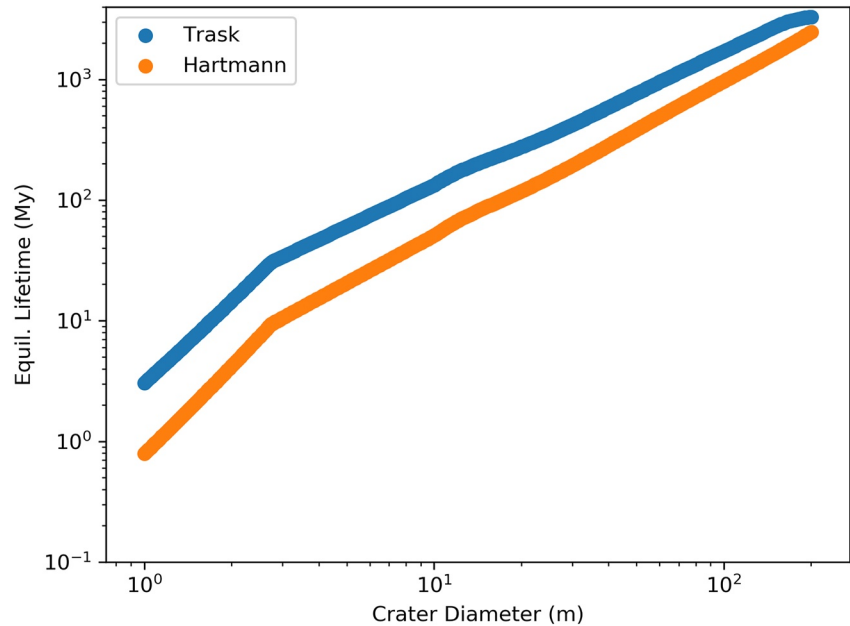


Figure 1. Lifetime of craters of different sizes assuming Trask equilibrium (Equation 3a) and Hartmann equilibrium (Equation 3b) with a crater production model from Neukum et al. (2001) above 10 m, scaled from Grün et al. (1985) model below 3.9 m, and interpolated between them in the 3.9–10 m size range (Figure S1 in Supporting Information S1). These assumptions require the lifetime of small craters ($D \leq 20$ –30 m) to be short, <500 Myr (see Table 1). The slopes on these equilibrium-derived lifetimes τ generally follow a power law with $\tau \sim D^{1.1}$ – $\tau \sim D^{1.3}$. The break in slope at ~ 2.7 m arises from the assumed change in the production function.

The cumulative density $\Phi(D, T)$ of craters formed $\geq D$ can then be expressed as:

$$\Phi(D, T) = C_{\geq 1\text{km}}(T) \Theta(D) \quad (8)$$

so

$$\Phi(D, T) = (5.44 \times 10^{-14} (e^{6.93T} - 1) + 8.38 \times 10^{-4} T) \left[(10^{\sum_{i=0}^{11} a_i (\log D)^i}) / 8.38 \times 10^{-4} \right] \quad (9)$$

or, for the period where the crater flux is inferred to be constant in the Neukum model, $T < \sim 2.5$ Ga, simply:

$$\Phi(D, T) = (10^{\sum_{i=0}^{11} a_i (\log D)^i}) T \quad (10)$$

To get the equilibrium lifetime $\tau(D)$ then only requires solving Equation 2 using the Neukum production model in Equation 9 (or the simplified version in Equation 10) and the equilibrium size-frequency distributions in Equation 3. One can think about this geometrically on a cumulative size-frequency distribution plot as the intersection points between the observed equilibrium curve and isochron at a particular size (Figure S2 in Supporting Information S1). Computing this intersection point is cumbersome analytically due to the Neukum production function's summed polynomial; however, it is easy to evaluate numerically. Results are given in Figure 1 and Table 1.

A wrinkle in this calculation is that, for diameters less than 10 m, the Neukum production model is undefined. To address the production rate of craters at 1–10 m sizes, we scale the impactor rate from Grün et al. (1985) to an equivalent cratering rate based on pi-scaling (see Table S2 and Figure S1 in Supporting Information S1). Grün et al. (1985)'s impactor flux is only constrained up to masses of 100 g, which is equivalent to $D \sim 2.7$ m crater with the scaling assumptions here. We thus also interpolate the production rates in the size range from $D \sim 2.7$ m to $D = 10$ m where the Neukum production rate is defined via linear interpolation on the log-log form of the production function. The Grün et al. production function has a steep, -4 cumulative form, which produces the steep break in slope in Figure 1 and Figure S1 in Supporting Information S1.

Along with an uncertain production function, there are also few measurements of the equilibrium density at sizes smaller than 10 m. One exception is Cadogan (2020), which included measurements of crater popula-

Table 1
Lifetimes in Millions of Years at a Given Size

Diameter (m)	Trask equilibrium crater lifetime (Myr)	Hartmann equilibrium crater lifetime (Myr)
1	3.1	0.8
1.41	6.5	1.8
2	14.5	4.2
2.83	31.6	9.7
4	46.9	15.3
5.66	69.6	24.1
8	103.3	37.9
10	133.3	50.8
11.31	157.9	61.5
16	224.7	92.8
20	275.9	118.4
22.63	311.4	136.5
32	453.0	210.6
40	588.2	284.0
45.25	681.3	335.9
64	1,028.2	537.7
80	1,331.6	723.3
90.51	1,533.0	850.4
128	2,273.5	1,337.9
160	2,915.3	1,805.1
181	3,184.5	2,142.7

Note. These lifetimes are most applicable to areas with minimal topography; craters on terrain with steeper slopes generally have shorter lifetimes.

tions down to $D \sim 2.5$ m using automated crater mapping on LROC NAC images. Cadogan's measurements in multiple locations fall between Trask and Hartmann equilibrium in Equation 3, supporting the approximate validity of these functions down to 1 m. There are also hints in Cadogan's data of a steeper power-law slope than expected in the equilibrium distribution at sub-10 m sizes, but this may be a result of scale-dependence in how effective automated detection was rather than a reflection of a bona fide shift in the equilibrium slope. Given the general consistency between Cadogan's results and Equation 3, it is likely reasonable to rely on Equation 3 to sub-10 m sizes as well. We note for completeness that Cadogan (2020) came to a similar conclusion of ~ 100 Ma lifetime for 10 m craters, consistent with what we find here.

4. Calculating Diffusivities Consistent With Equilibrium Lifetimes

From the crater lifetimes given in Figure 1 and Table 1, it is straightforward to directly infer the effective diffusivity κ at a given size with two additional inputs: (a) the shapes of fresh craters when they form on the Moon; and (b) the threshold for when craters become unrecognizable. With these inputs, we can simply take the initial shape, apply the diffusion model until the visibility threshold at some degradation state $\kappa\tau$ is attained, and then divide by the crater lifetime τ at that size.

When making this calculation, it is important to note that neither the initial shape nor the visibility threshold is known with certainty, especially in the meter-to-decameter size range. We make some reasonable assumptions based on existing data. For fresh crater shape, we first tried using the fresh crater profiles in FT2014, extrapolated to smaller scale. However, observations from LROC DTMs show that small fresh craters ($D \sim 30$ m and below) on the Moon rarely have as high depth/Diameter (d/D) values as their larger counterparts, rather, they show a decline in d/D with decreasing crater size (Mahanti et al., 2018; Stopar et al., 2017; Sun et al., 2018). This is also very evident when making synthetic d/D populations and comparing them to

observations (see Section 5). Thus, as our baseline, we assume a declining $(d/D)_{\text{initial}}$ with crater size for fresh craters (Table 2) that is then expressed in their initial shapes (Table 3).

The values in Table 2 capture only a characteristic value for the fresh crater depth at a given size. Observations suggest that there is some natural variability that may arise as a function of impactor or target properties. For example, we do not attempt to capture more complicated profiles observed in some small lunar craters, like mounds or concentric terraces (Bart et al., 2011; Quaide & Oberbeck, 1968; Yang et al., 2021). Yang et al. (2021) show that these morphological differences are rapidly modified away as degradation proceeds, so this simplification likely has little effect.

For the visibility threshold, we chose to use a single cutoff value for d/D , which is similar to the approach of Minton et al. (2019). We selected $(d/D)_{\text{cutoff}} = 0.04$ as the nominal case based on the left tail of the probability density functions for d/D in Mahanti et al. (2018). Choice of the $(d/D)_{\text{cutoff}}$ has an appreciable effect on the absolute diffusivities, but not on the qualitative behavior or relative diffusivities at different scales. Higher visibility thresholds need lower diffusivities and vice versa because less diffusion is needed to erase craters. In reality, it is conceivable that the visibility threshold is size-dependent, or the $(d/D)_{\text{cutoff}}$ in practice varies with size. This is not easy to evaluate, as the threshold may depend strongly on measurement methods (Robbins et al., 2017), illumination geometry (e.g., Richardson et al., 2022;

Table 2
Assumed $(d/D)_{\text{initial}}$ at Various Sizes

Diameter (m)	Initial d/D (rim to floor)
≤ 10	0.1
10–40	0.11
40–100	0.13
100–200	0.15
200–400	0.17
≥ 400	0.21

Note. The bottom four rows here (>40 m) are based on Stopar et al. (2017). The smallest two diameter bins are extrapolated and uncertain given existing data on the freshest craters in these size ranges.

Table 3

Implementation of Fresh Crater Model (Modified After FT2014; Primary Change Is in Handling of Interior Floor and Initial d/D)

	Distance from crater center R normalized by diameter, $R = r/D$ ($R = 0$, center; $R = 1.0$, Rim)	Formula
Interior on wall	$R < 0.98$, where $Z > -((d/D)_{\text{initial}} - 0.0368)D$	$z = (-0.229 + 0.227R + 0.083R^2 - 0.394R^3)D$
Interior on floor	$R < 0.98$, where $Z \leq -((d/D)_{\text{initial}} - 0.0368)D$	$Z = ((d/D)_{\text{initial}} - 0.0368)D$
Rim	$R = 0.98-1.02$	$z = 0.0368D$
Exterior	$R = 1.02-1.5$	$Z = (0.188 - 0.187R + 0.018R^2 + 0.015R^3)D$

Wilcox et al., 2005), and other factors affecting crater mappers' individual thresholds for crater recognition (Robbins et al., 2014).

By taking the initial shape and diffusing craters until they reach the assumed visibility threshold, we calculate an estimate for effective diffusivity κ at different scales (Figure 2) over craters' lifetimes (Figure 1, Table 1). The results of this calculation show that effective diffusivity must decrease as craters decrease in size, implying that it does so for topography with shorter length-scales as well.

The step discontinuities in Figure 2 are a result of the changes in the presumed crater shape. The transition in slope at $D \sim 3$ m is at the size where the assumed production function changes for the assumed scaling parameters (see Figure S3 in Supporting Information S1). A factor of 2–4 $\times$ uncertainty in the effective diffusivities is implied by the different equilibrium functions alone (Trask vs. Hartmann), with additional uncertainties from the initial morphometry (see Figure S4 in Supporting Information S1), crater production function, and the visibility threshold. Below $D = 10$ m, the uncertainty in the production function adds appreciable additional uncertainty (perhaps another factor of 2 $\times$). The results here are still sufficient to improve qualitative understanding and for making tentative quantitative inferences about surface evolution at different scales.

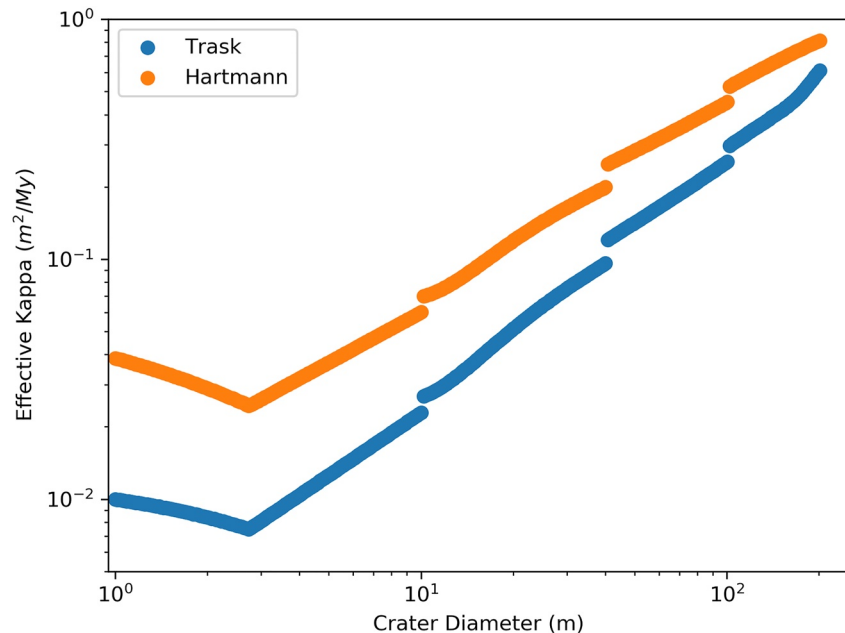


Figure 2. Effective diffusivity required for maintaining crater equilibrium over the timescales given in Figure 1 and Table 1, assuming the crater morphometry model from Tables 2 and 3 and a visibility threshold of $(d/D)_{\text{cutoff}} = 0.04$. Trask equilibrium (blue; Equation 3a) and Hartmann equilibrium (orange; Equation 3b) differ here because the Hartmann lifetimes are shorter (Figure 1, Table 1). Above 5-m, the effective diffusivity implied by the Hartmann equilibrium function can be approximated by a power-law $\kappa_{\text{eff}} \approx 3.6 D^{0.87}$, and the Trask diffusivity has $\kappa_{\text{eff}} \approx 3.1 D^{1.05}$ for D in km and κ_{eff} in m^2/Myr .

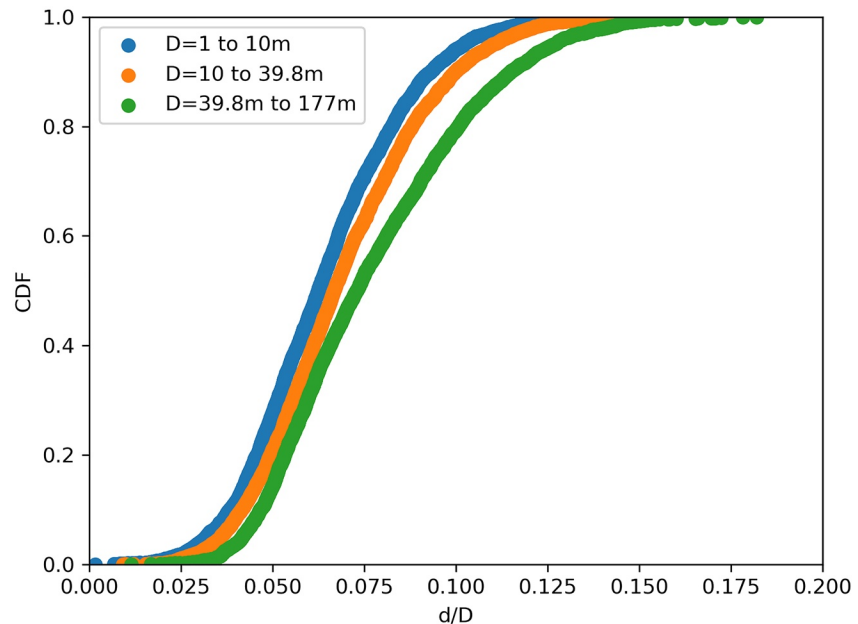


Figure 3. Cumulative density functions of the d/D (depth/Diameter) values produced by synthetically diffusing craters in equilibrium. Note that this calculation was made with different surface areas A for expediency ($A_{1-10} = 0.05 \text{ km}^2$, $A_{10-39.8} = 5 \text{ km}^2$, $A_{39.8-177} = 80.0 \text{ km}^2$), which is useful to get a statistically meaningful sample of craters at different sizes in a reasonable amount of computation time.

5. Synthetic Populations of Craters in Equilibrium

It is useful to have a sense of the depth-frequency distribution of craters in equilibrium on the Moon for mission planning and hazard assessment. Such distributions are easy to synthesize from the assumptions described above, given (a) initial shapes of fresh craters (Tables 2 and 3) (applying a standard deviation of $\sigma = 0.02$ variability to the initial d/D), (b) a visibility/removal threshold (assuming $(d/D)_{\text{cutoff}} = 0.04$, as above), (c) an equilibrium size-frequency distribution (Equation 3), and (d) effective diffusivities κ that are consistent with the observed equilibrium size-frequency distribution (Figure 2). Taking these together, we can produce synthetic d/D distributions as a function of crater size (Figure 3).

The results here are largely dependent on the fresh crater morphometry and the visibility threshold (a, b) and not the equilibrium density or effective diffusivities (c, d). The reason is that crater degradation states extend from fresh to almost erased regardless of the effective diffusivity or equilibrium size-frequency distribution, and they degrade most quickly when fresh and then less quickly approaching removal (see also Basilevsky, 1976; Basilevsky et al., 2014; FT2014; Mahanti et al., 2018). The form of the observed d/D distribution is highly dependent on what size-range is considered because of variability in the typical initial crater shape with crater size (i.e., Table 2).

Observations are consistent with this inference that most lunar craters are topographically degraded rather than fresh (i.e., comparing Figure 3 to Mahanti et al. (2018)'s Figure 11b (plains curves) or to Figure 10 from Basilevsky et al., 2014). The density functions for d/D are a slightly right-skewed distribution with a pronounced tail for the freshest craters. The distribution of d/D values in any particular area of the lunar surface should be expected to differ from these simulations due to the vagaries of the local cratering history, contributions of secondaries, and due to any reduction in crater lifetime due to regional slopes. Other factors not considered here may also play some role in local d/D distributions, either by affecting degradation rates or fresh crater shapes, such as differences in geologic materials, recent tectonic activity, or distinctive thermal environments.

6. Discussion

6.1. Applicability, Scale/Time Confusion, and Other Uncertainties

A few comments on the applicability of this paper are worth discussing. One key limit of this analysis is that it only applies to areas with low background slopes. From past measurements, it is well-established that crater lifetimes are greatly reduced in areas of appreciable slope and the resulting equilibrium densities are also reduced. Consequently, the lifetimes given here are likely upper bounds on crater lifetimes if considering areas with higher background slopes.

Additionally, we focus our derivation of an equilibrium constraint on effective diffusivity to craters 1–200 m in diameter. The reason we do not extend the analysis to craters bigger than 200 m is that craters larger than that size have not reached equilibrium over the age of typical maria (i.e., since the Imbrian) (e.g., Xiao & Werner, 2015). Early in the Imbrian and before, the effective diffusivity was likely greatly enhanced simply due to the higher impact flux characteristic of early lunar history, so untangling differences in diffusivity as a function of scale from those related to time becomes increasingly intractable.

Indeed, a broad complication for the effective diffusivities computed here is that they functionally sample different time periods. This issue arises because craters of distinct sizes have such different lifetimes (i.e., 1 m craters only sample a few million years, given their lifetime, whereas 20 m craters sample ~120–275 Myr; see Table 1). Because we assume a constant diffusivity over the course of any craters' lifetime, any changes in diffusivity with time (independent of scale) would end up expressed in these effective diffusivities as a scale-related effect. In other words, Figure 2's calculation implicitly assumes no temporal changes in effective diffusivity independent of the presumed crater flux. Any temporal changes in the cratering rate that was not captured in the Neukum cratering chronology applied here would also be obscured (i.e., Hartmann et al., 2007; Mazrouei et al., 2019; Quantin et al., 2007).

FT2014's results suggest that this assumed temporal consistency in diffusivity is approximately true for the last 3 Ga when the period that the cratering rate is treated as constant; the measurements in FT2014 do not suffer from a similar convolution of spatial and temporal changes, as the craters in FT2014 were close to the same size (96% were between $D = 0.8$ and 2 km in diameter). Because the changes in effective diffusivity as a function of scale are quite large, nearly proportional to D , any time evolution of diffusivity is unlikely to overwhelm the scale-dependent effect.

There are two other known uncertain elements of our approach and results that are worth discussing. First, the effective diffusivity in Figure 2 in the Grün-controlled part of the curve below ~3 m in diameter is somewhat anomalous. Above that size, effective diffusivity increases with crater size, consistent with the discussion in Section 2 that as craters get larger, they are also exposed to degradation by processes linked to more sizable craters and we should expect they have higher effective diffusivities. The fact that the sub-3m effective diffusivities start to increase as sizes decrease is thus highly suspicious and a likely consequence of the assumed production function, equilibrium function, visibility threshold, and/or fresh crater morphometry. Unfortunately, all of these factors are badly known at sub-3m sizes. More work is needed to improve the calculations presented here in the meter-scale size range, but the form of the results calculated here are more likely to be reasonable >3 m than below this size.

Second, as craters degrade and infill, they also widen (FT2014; Bugiolacchi & Wöhler, 2020; Xie et al., 2017) as the rim degrades inward and is eventually destroyed. For analytical tractability, all our calculations neglect this effect. For example, the production rate of craters (e.g., Equation 5) is treated as the crater production rate at a fixed size, but in practice, widening pushes craters to larger sizes as they degrade. This tends to increase the effective production at a given size as more frequently formed small craters widen into that size. Likewise, the degradation needed to erase craters at a given size was also calculated without reference to widening. Taking into account widening would drop the needed diffusivity at a given initial size, because d/D would decline more quickly. These effects on production and erasure thus at least partially counteract each other.

Accurate consideration of this widening effect is not simple, because it is also convolved with how the crater production function is defined, how craters are mapped, and the threshold when they become invisible. The NPF as originally measured was built by cataloging the population of craters on surfaces of some age, so implicitly already captures craters in a range of degradation states and evolved widths, rather than being a true fresh crater

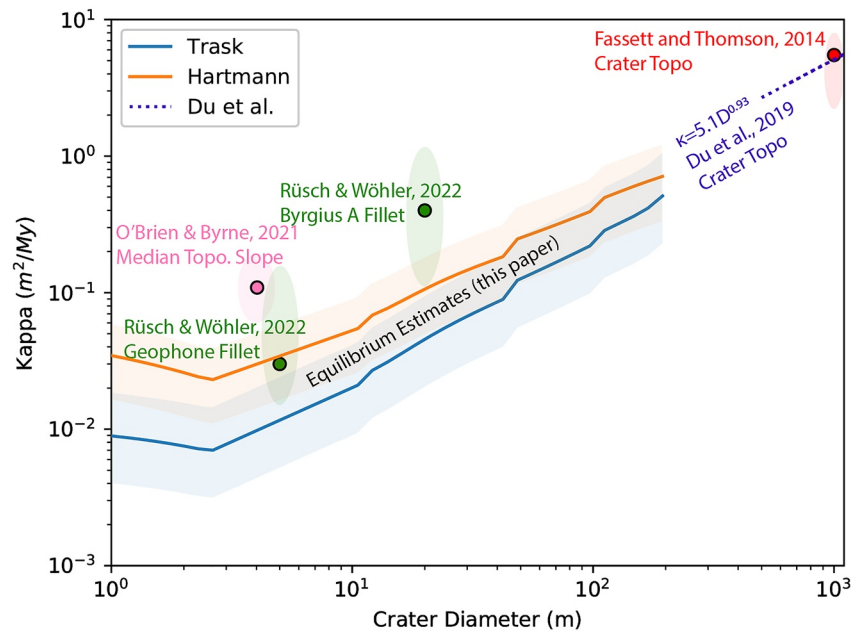


Figure 4. Comparison of the effective diffusivity required for maintaining crater equilibrium at different scales in this work to earlier literature estimates. Note that there are factors-of-a-few uncertainties on the Trask and Hartmann curves due to the visibility threshold, production function, and crater shape assumptions, and significant uncertainties in the literature values as well. All uncertainties depicted here (both this study and the literature values) are schematic rather than from formal error analysis.

production function. Widening of craters from the NPF to produce an exposure-dependent production function is not exactly self-consistent.

We expect that the errors in the effective diffusivity estimates from neglecting crater widening are less than a factor of $2\times$ at a given size, so smaller than the uncertainties from the different equilibrium functions, but comparable to other uncertainties caused by morphometry or visibility parameterization.

6.2. Comparisons to Earlier Diffusivity and Lifetime Estimates

We compare the results given in Figure 2 with several earlier literature estimates for the diffusivity in Figure 4. For example, if one were to extrapolate the equilibrium function-derived effective diffusivities derived here to the crater sizes considered in FT2014, the result is $2.8\text{--}3.2\text{ m}^2/\text{Myr}$ (see also Figure 2), while FT2014 estimated $5.5\text{ m}^2/\text{Myr}$ for km-scale craters. In our view, the diffusivity at km-scale estimated by FT2014 from crater degradation and the results here agree reasonably well, given the extrapolation, underlying uncertainties, and the possibility of scale/time confusion. For example, it is plausible that the effective diffusivity somewhat steepens between the maximum crater size constrained by equilibrium ($\sim 150\text{--}200\text{ m}$; see above) and kilometer-sizes, as larger craters are deeper and thus maintain steeper slopes longer than those at smaller scales. FT2014's kilometer-scale diffusivity is essentially consistent with a more recent measurement at that same scale and large by Du et al. (2019), who also quote a scale dependence for diffusivity with power-law exponent 0.93, consistent with results both here and in Fassett et al., 2018.

Another recent study addressing effective diffusivity looked at the shape of regolith piles that form piled up next to boulders (so-called rock fillets) (Rüsch & Wöhler, 2022). The detailed morphometry of fillets that form adjacent to boulders are sensitive to the rate at which regolith is produced and diffused away from rocks as they weather. Since some individual rocks at landing sites have ages estimated using exposure-age dating, a local estimate for regolith diffusivity is possible from fillet morphometry. Rüsch and Wöhler (2022) quote an order of magnitude uncertainty on the resulting diffusivities. Combined with the uncertainties here and the fact that fillet diffusivity might diverge for process reasons somewhat from diffusivity inside craters, Rüsch and Wöhler's rock fillet results overlap reasonably with our estimates of diffusivity.

In a comprehensive study of mare terrain evolution, O'Brien and Byrne (2021) derived a diffusivity estimate by evaluating the median pixel-to-pixel slope on a 4 m/px model grid in their numerical landform evolution model and comparing it to the median slopes at the same scale at the Apollo landing sites. From this comparison, they derive a required diffusivity of $0.109 \text{ m}^2/\text{Myr}$ at 4 m scale, a factor of $\sim 4\times$ higher than the Hartmann equilibrium implied value we calculate. However, the magnitude of the disagreement between their work and ours is again somewhat less than it first appears because all fresh craters in their model have a d/D ratio of 0.2 rather than the d/D of 0.1 for fresh craters we are assuming at this same scale.

Qualitatively, the results in this paper are likely within the mutual uncertainties of past studies. Further measurements that better bound the observed small-scale equilibrium behavior and effective diffusivities would be valuable, as would obtaining additional absolute ages for craters and surfaces on the Moon.

The crater lifetimes we report here are much longer in general than calculated by Bugiolacchi and Wöhler (2020), and both Bugiolacchi and Wöhler (2020) and Zhang et al. (2021) found much faster diffusivities at 10 and 100 m than derived here. We note this is not caused by their inclusion of crater widening; as we discuss above, we expect this should reduce the needed effective diffusivities, rather than raise them. Reconciling these results are beyond the scope of what we attempt here.

6.3. Crater Lifetime and Volatile Exploration

It has been well-established that impact gardening is an important process affecting the removal and burial of volatiles at the lunar surface (e.g., Costello et al., 2020; Crider & Vondrak, 2003; Farrell et al., 2019; Hurley et al., 2012) that are likely found in Permanently Shadowed Regions (PSRs). The short lifetimes for small craters in Figure 1 and Table 1 are quite consistent with impact gardening models that suggest a rapid churning of the near-subsurface, implying volatiles in the upper tens of centimeters must be emplaced (geologically) recently. Indeed, the crater lifetimes we present here give an additional perspective on volatiles emplaced in crater-hosted PSRs that is complementary to that of Monte Carlo models of impact gardening.

Costello et al. (2020) estimated that micro-PSRs (e.g., Hayne et al., 2021) with depths of less than 30 cm are $<10 \text{ Myr}$, based on their inferred gardening rate. Our results are similar but are best understood as bounding the maximum age of volatiles found within recognizable craters of specific sizes, at least assuming those volatiles were deposited after the formation of the crater and were not pre-existing and somehow survived during the impact event. This assumption of post-crater deposition seems likely given the amount of heat and energy available to devolatilize following impact events.

Because it is easier to maintain deep shadows in fresher, deeper craters, PSRs are more common in relatively fresh craters, especially at small scales (e.g., Hayne et al., 2021). Thus, the upper limits we quote above are likely to be well above the typical age of ice-hosting PSRs found at these sizes. If the assumption is correct that volatiles were emplaced after small craters, and the morphometry of volatile-hosting craters can be determined in situ, direct estimates for ages (rather than maxima) for the timing of ice deposition may be determined.

7. Conclusion

In this manuscript, we show how the known equilibrium size-frequency distribution of craters is directly connected to crater destruction due to diffusive infilling. With a known production rate combined with the equilibrium size-distribution, how long craters survive as recognizable on the lunar surface can be directly calculated. As has been recognized before, these lifetimes, especially for small craters, are geologically short (Figure 1 and Table 1). The resulting effective diffusivities at various crater sizes are shown in Figures 2 and 4: these vary by orders of magnitude at different scales, consistent with anomalous diffusion.

The crater degradation model we revisit here has wide ranging implications for lunar surface evolution. For example, we show how this degradation model can be used to predict, and successfully match existing observations, of the statistical distributions of d/D for craters at scales relevant to lunar surface exploration (Figure 3). Additionally, the short lifetime of crater at small-scales implies that if ice is found in small crater-hosted micro-PSRs and was delivered after their formation, it was likely a recent event or process. This will be an exciting target for exploration by the Volatiles Investigating Polar Exploration Rover mission (Colaprete et al., 2021).

Data Availability Statement

A released version of the code used for this paper is archived at Zenodo (C. Fassett, 2022), or may be available on github at: <https://github.com/cfassett/DiffusiveEquilibrium>.

Acknowledgments

Support for CIF's work on this topic comes from the KPLO PS program, the VIPER mission, and past support from the LDAP program. Minggang Xie and an anonymous reviewer provided constructive and helpful peer review comments that helped improve the manuscript.

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